

Training Configuration and Summary:

Dataset used: rssi_odom_dataset_12apr25_ajust.parquet

model_type: LSTM
group_sec: 1
timesteps_orig: 30
timesteps: 30
batch_size: 256
min_units: 32
max_units: 256
units_step: 32
dropout_min: 0.2
dropout_max: 0.3
dropout_step: 0.1
activation_functions: ['tanh', 'relu', 'elu']
optimizers: ['adam', 'rmsprop', 'Nadam']
min_layers: 1
max_layers: 2
max_epochs: 150
hyperband_iterations: 3
forecast_steps: 20

LSTM Multi-Step Training Report

Report Generated: 2025-05-09 23:31:54

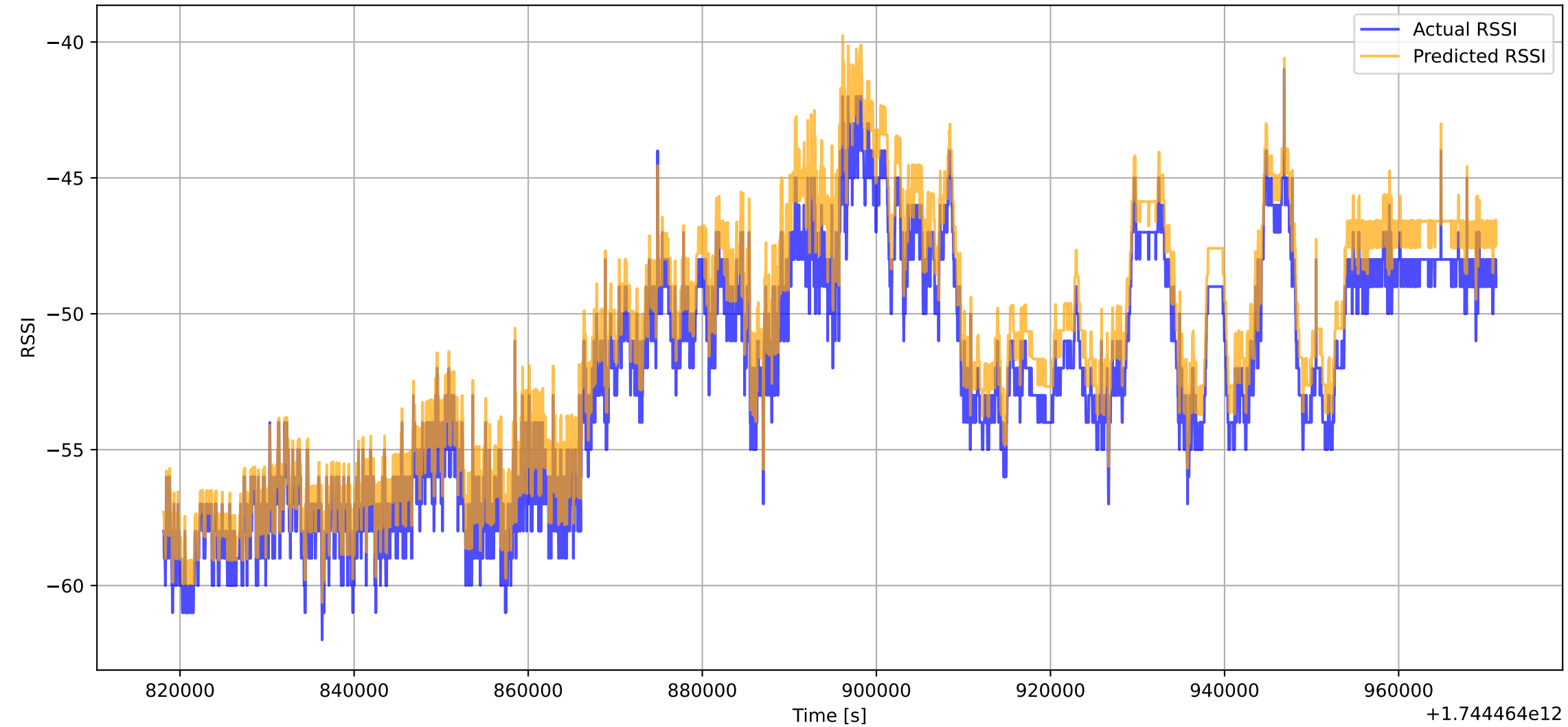
Training Start: 2025-05-09_16-13-36
Training End: 2025-05-09_17-00-59
Training Duration: 47.38 minutes

Best Hyperparameters:
Number of Layers: 1
Activation: tanh
Optimizer: Nadam
Dropout Rate: 0.20

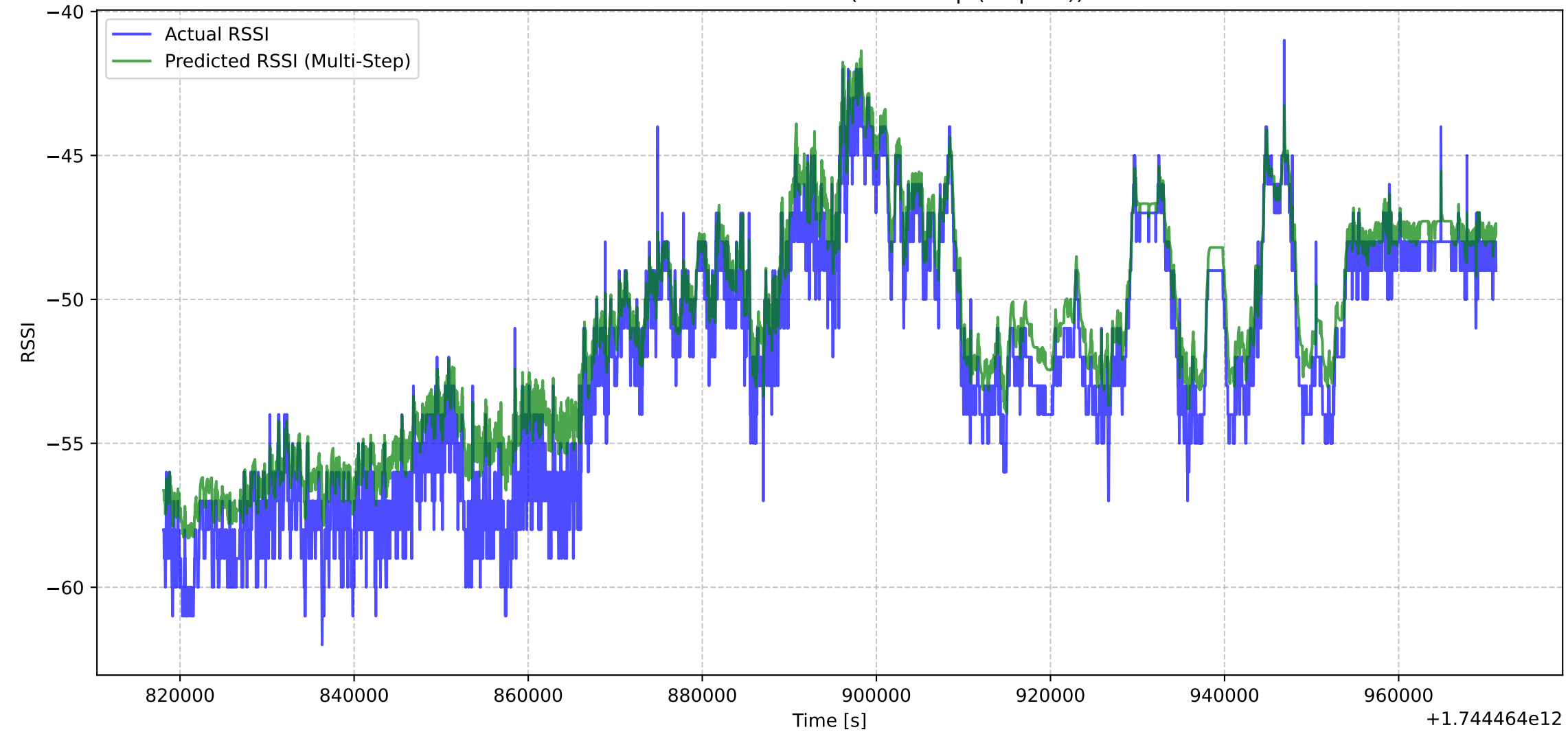
MAE: 1.4062
MSE: 2.8708
RMSE: 1.6943
R2 Score: 0.8417

Multi-Step Forecast:
Forecast Steps: 20
MAE: 0.0192
MSE: 0.0004
RMSE: 0.0212
R2 Score: 0.8893

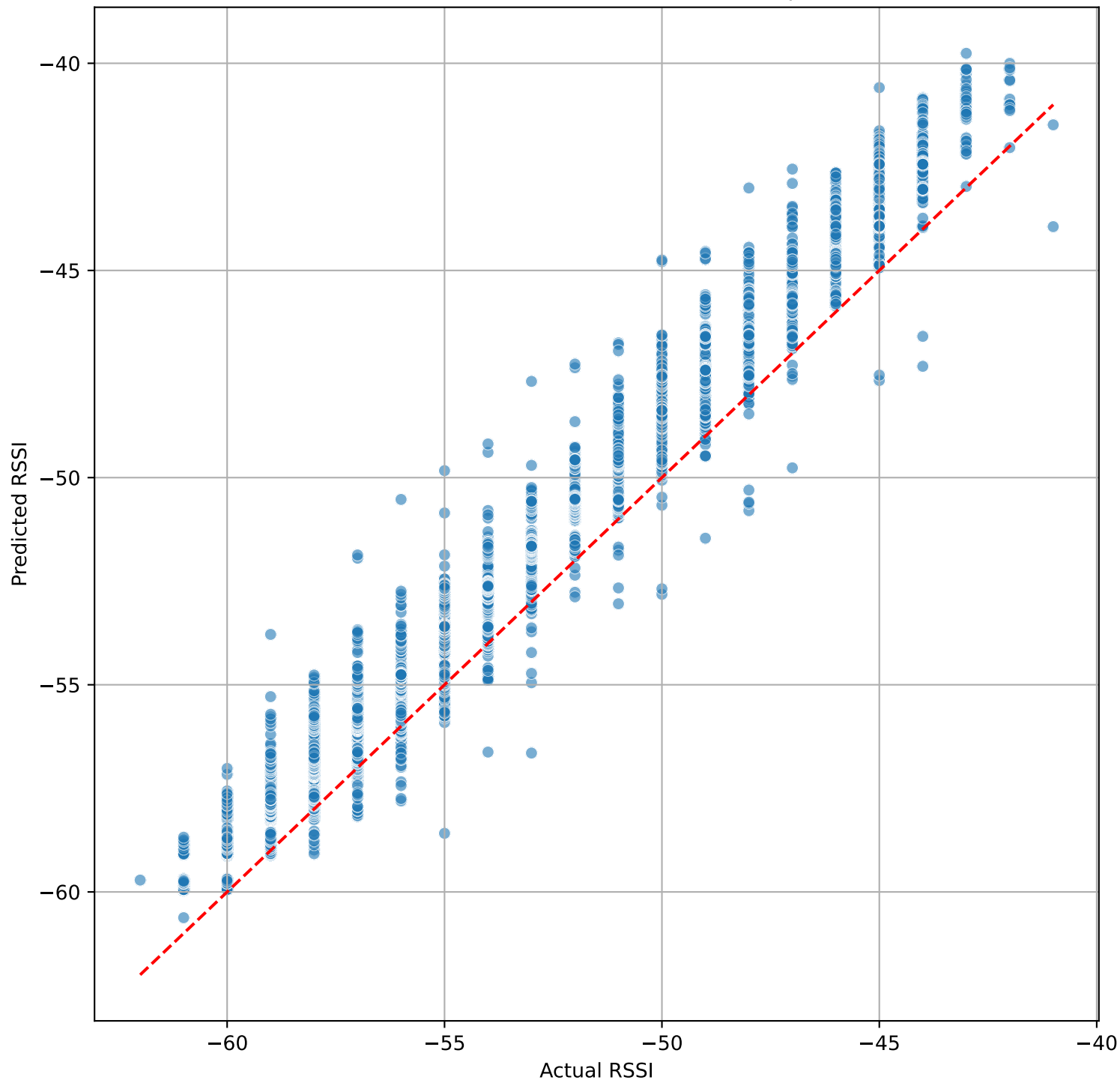
Actual vs Predicted RSSI (First Step)



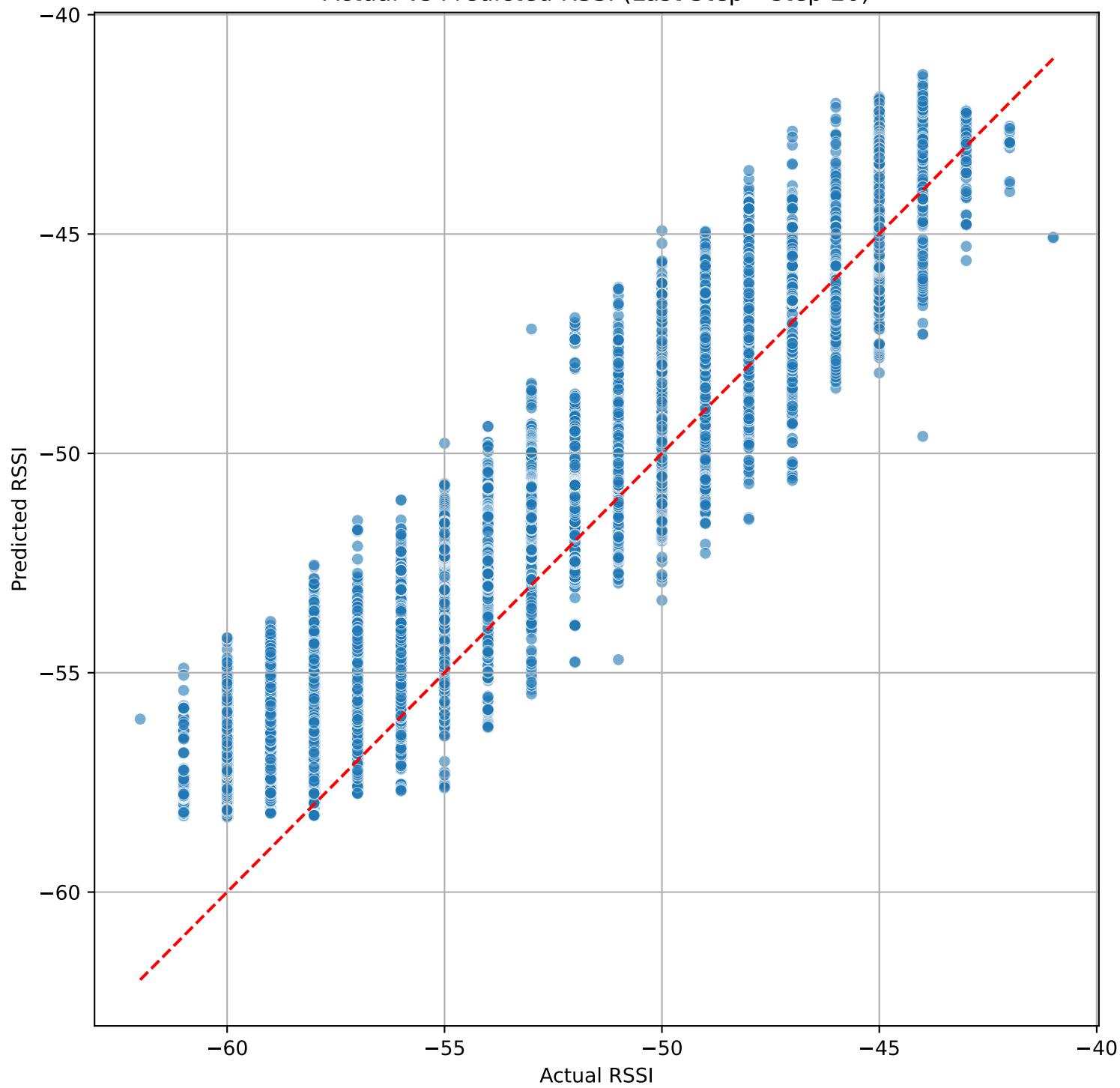
Actual vs Predicted RSSI (Multi-Step (step 20))



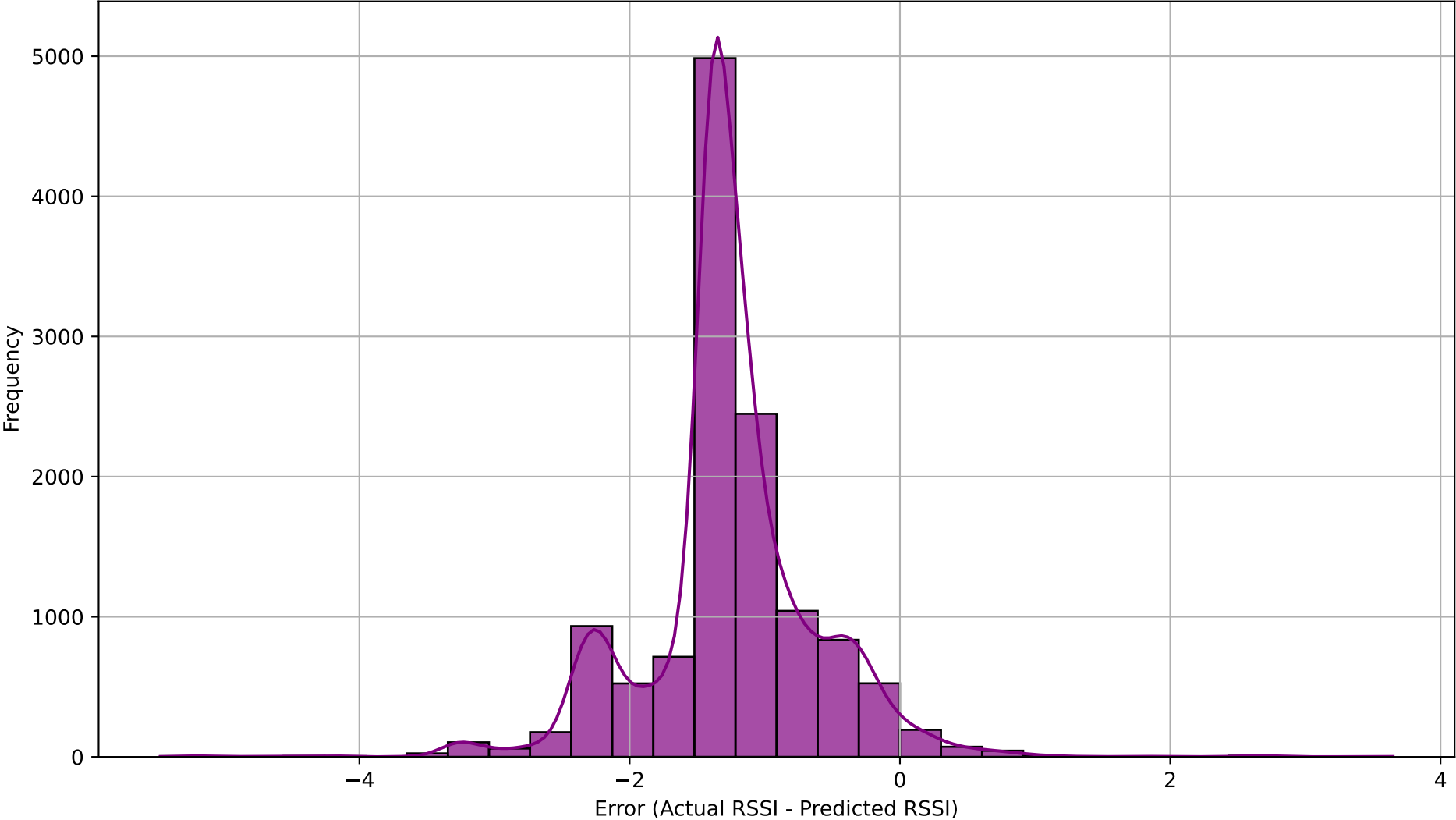
Actual vs Predicted RSSI (First Step)



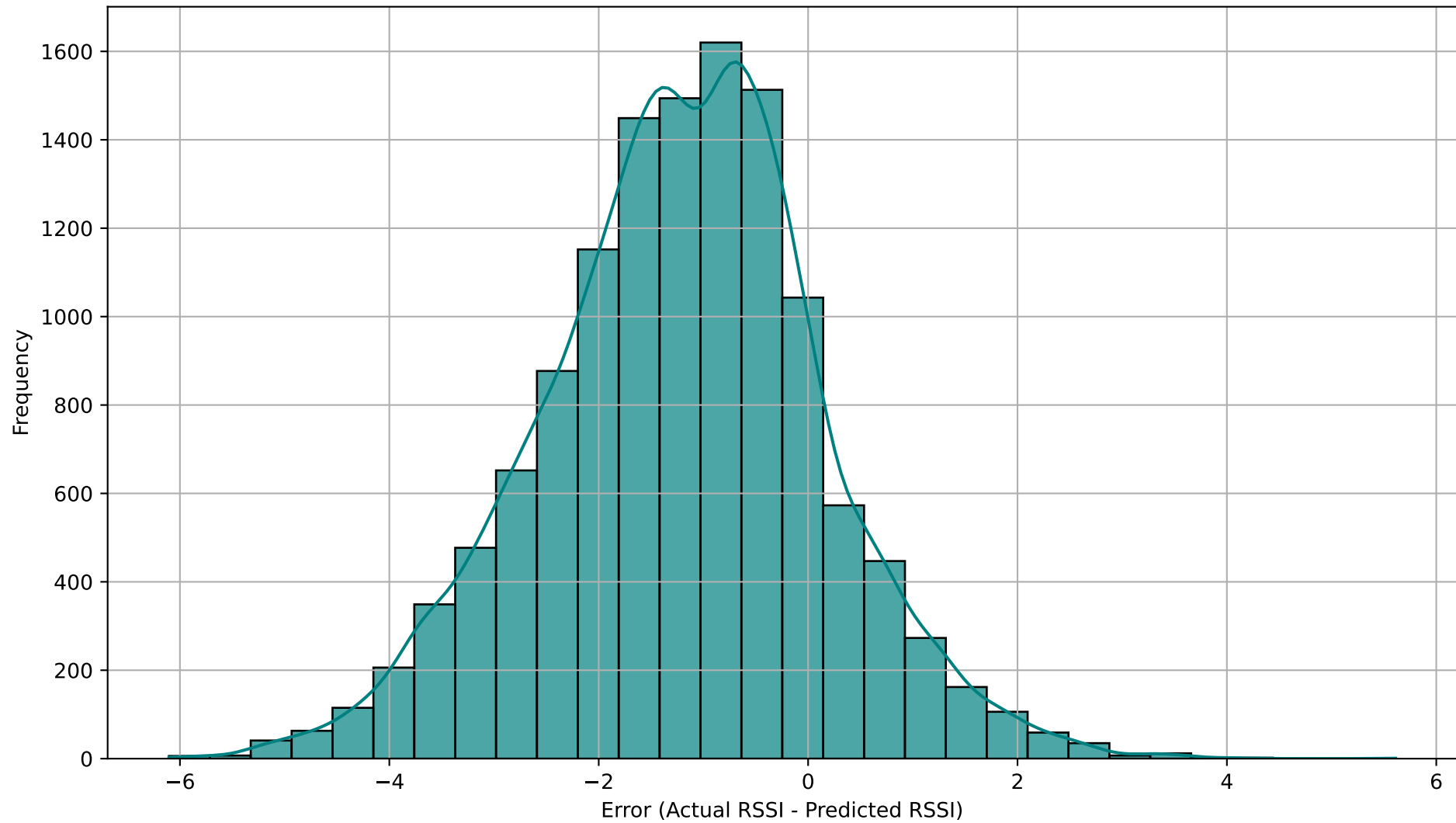
Actual vs Predicted RSSI (Last Step - step 20)



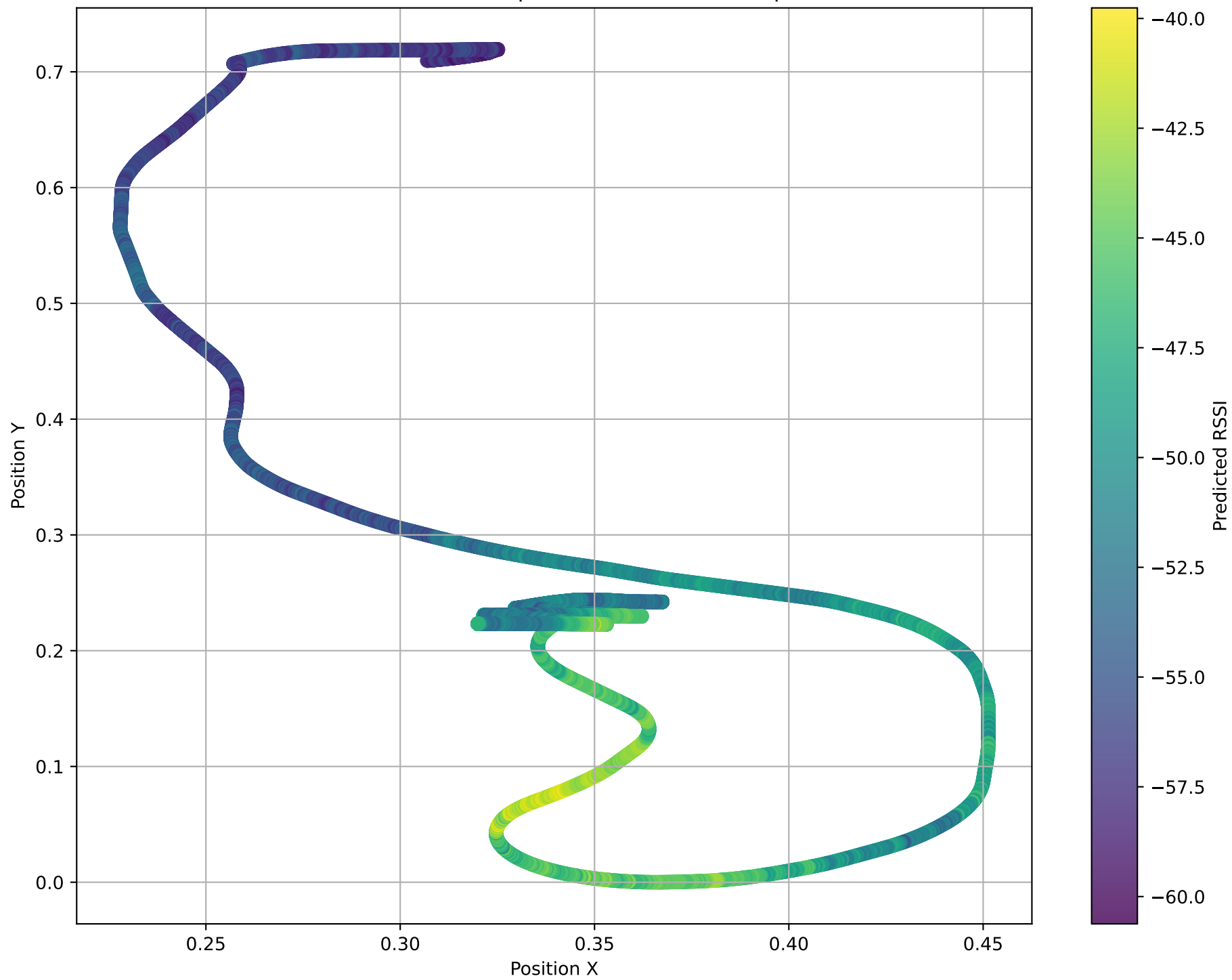
Distribution of Prediction Errors (First Step)



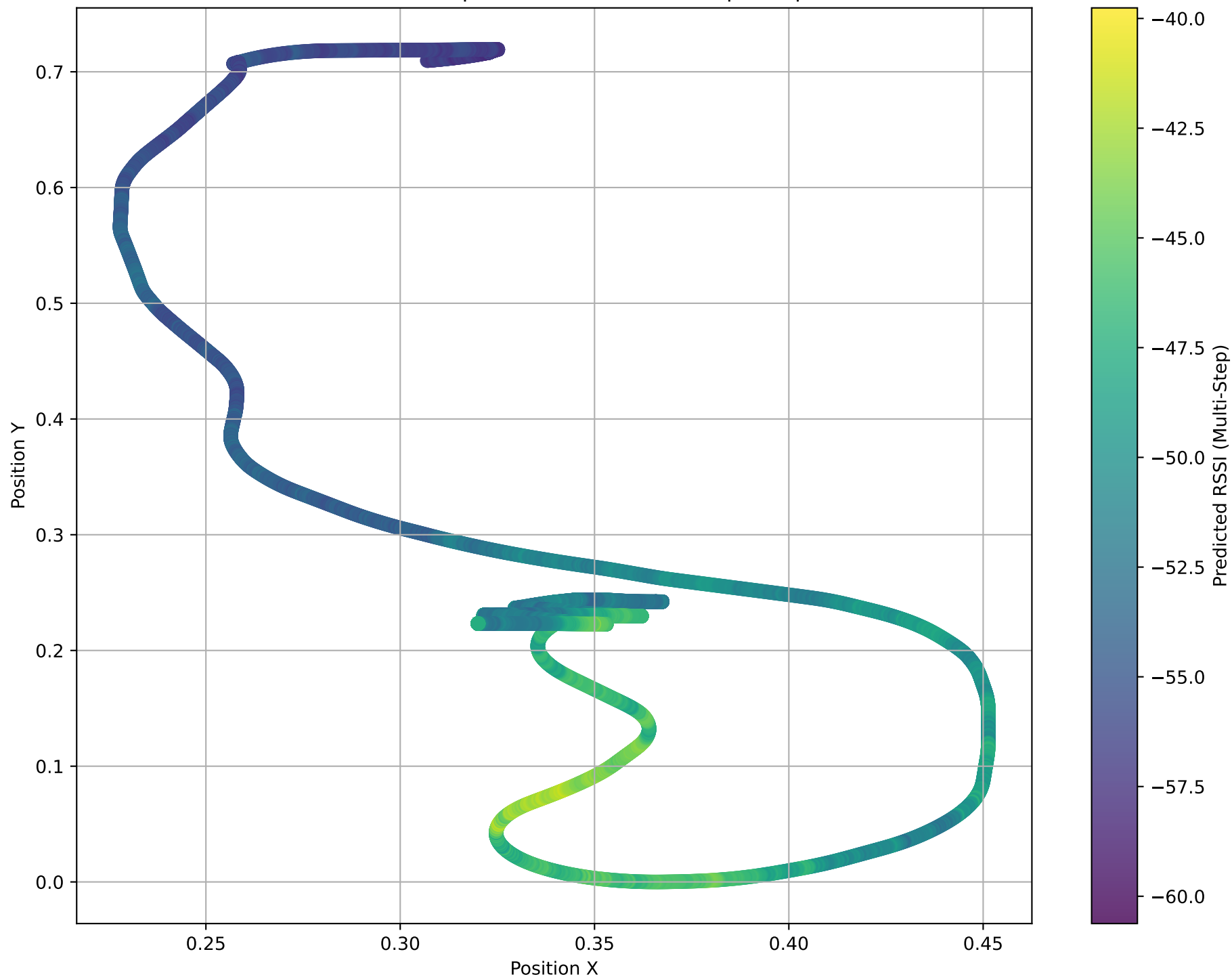
Distribution of Prediction Errors (Last Step - step 20)



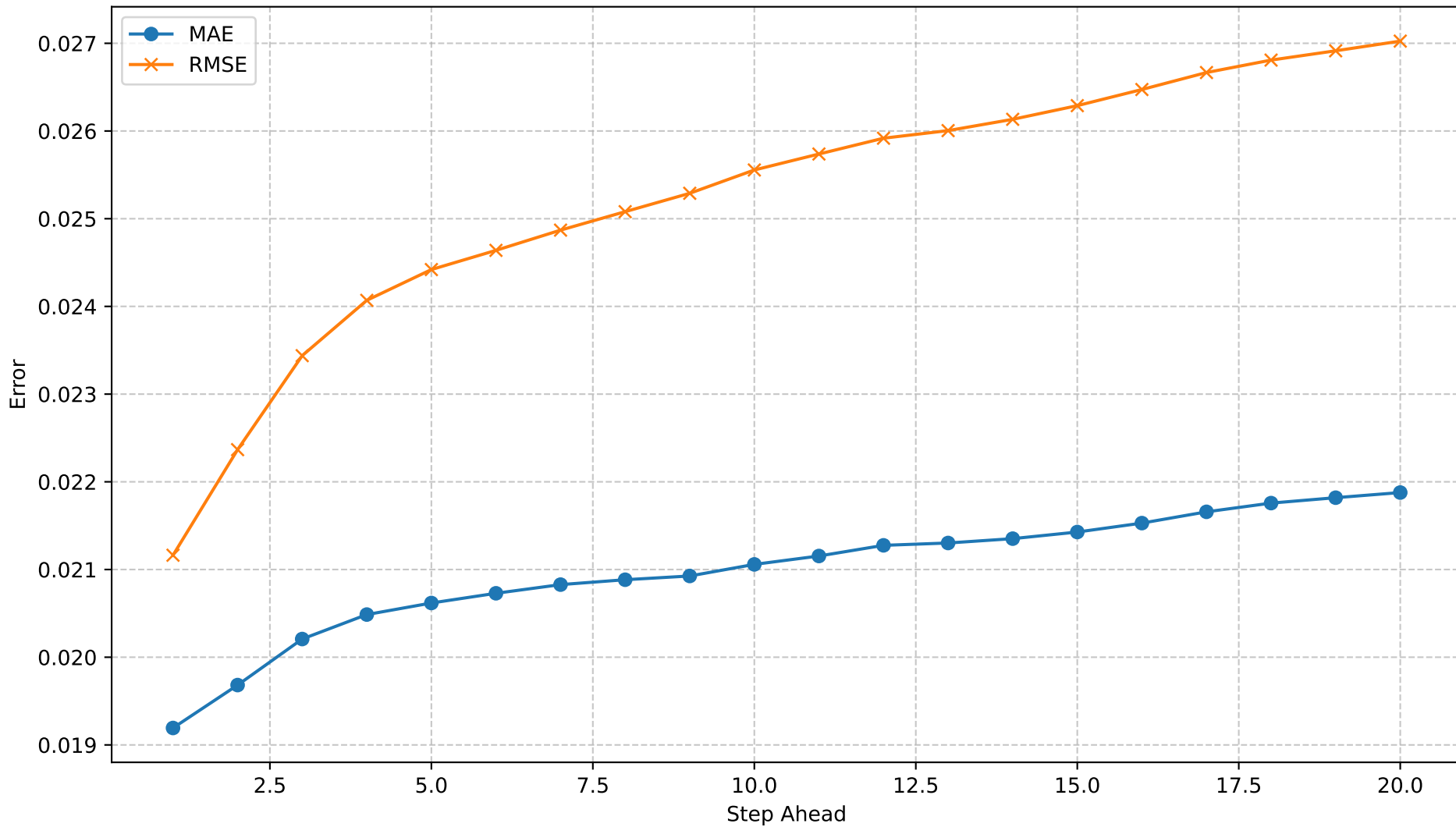
Predicted RSSI in Spatial Context (First Step)



Predicted RSSI in Spatial Context (Multi-Step - step 20)



Multi-Step Forecast Error by Step



Resumo Automático

Dataset: rssi_odom_dataset_12apr25_ajust.parquet

Modelo: LSTM

Forecast steps: 20

MAE / RMSE (dB)

- 1º passo : 0.02 / 0.02
- Mediana : 0.02 / 0.03
- 20º passo : 0.02 / 0.03

Crescimento do erro: 14.0% → Estável (erro cresce < 20 %)

3 piores passos (MAE): 20 (0.02), 19 (0.02), 18 (0.02)

Recomendação:

0 modelo é adequado para decisões de hand-over até o horizonte máximo configurado.